

POPLAR BLUFF, MO, USA

WMO: 723300

Lat: 36.773N

Lon: 90.325W

Elev: 327

StdP: 14.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	18.0	0.0	5.5	16.2	4.9	7.2	20.8	24.6	47.4	21.7	47.5	6.4	320	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	94.0	77.5	92.0	76.9	90.3	76.3	80.7	89.9	79.4	88.7	78.2	87.0	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	149.7	86.2	76.9	141.6	84.7	75.4	134.6	83.2	44.5	90.3	43.1	88.7	41.8	87.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	17.9	15.9	DB	6.3	97.9	5.0	2.7	99.9	-0.2	101.5	-3.1	103.0	-6.7	105.0		
			WB	4.5	82.7	4.9	1.6	83.9	-1.8	84.8	-4.6	85.6	-8.1	86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	36.1	40.1	48.8	59.8	68.5	76.8	79.4	78.0	71.0	60.0	48.3	39.2
	DBStd	17.35	10.53	10.71	9.95	8.13	7.27	5.05	4.49	4.45	6.41	8.32	8.79	9.62
	HDD50	1421	448	308	145	16	1	0	0	0	0	13	135	355
	HDD65	3875	897	698	508	193	52	1	0	0	21	201	504	800
	CDD50	4676	16	30	107	310	574	805	912	867	630	322	82	21
	CDD65	1655	0	0	5	37	160	356	447	402	201	45	2	0
	CDH74	15938	0	1	38	302	1350	3612	4556	3779	1812	465	23	0
	CDH80	6407	0	0	2	47	394	1534	2004	1629	680	116	1	0
Wind	WSAvg	6.3	7.4	7.7	8.0	8.1	6.7	5.6	4.8	4.3	4.3	5.3	6.4	6.7
Precipitation	PrecAvg	47.8	3.4	3.1	4.4	4.9	5.0	3.7	4.1	3.3	3.3	3.5	4.7	4.3
	PrecMax	65.0	6.2	8.0	11.6	16.6	10.6	7.3	8.4	11.6	6.7	13.5	12.5	9.7
	PrecMin	26.5	1.1	1.1	2.1	0.0	0.0	1.0	0.9	0.5	0.5	0.0	1.2	0.7
	PrecStd	8.6	1.6	1.5	2.1	3.1	2.2	1.7	2.0	2.2	1.7	2.7	2.9	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	71.4	78.6	84.2	90.8	96.4	97.3	96.9	93.4	88.0	77.7	67.7
		MCWB	61.3	63.0	64.5	67.4	73.0	76.8	79.4	79.1	75.2	71.5	64.2	62.4
	2%	DB	63.0	65.9	73.7	81.1	87.6	92.9	93.7	93.3	90.4	83.0	71.7	63.2
		MCWB	59.0	58.3	63.1	66.5	72.3	76.0	78.7	77.5	74.7	68.3	60.8	58.5
	5%	DB	57.5	62.2	69.3	77.6	84.3	90.7	91.5	91.0	87.6	79.5	67.4	58.8
		MCWB	52.6	55.2	59.4	64.1	70.6	75.1	78.2	77.2	73.4	65.6	59.9	54.3
	10%	DB	52.7	57.4	65.0	74.0	81.5	88.4	89.8	88.5	83.9	75.1	63.7	54.4
		MCWB	47.9	51.5	56.6	62.7	69.4	74.3	77.8	76.2	71.3	63.6	56.9	50.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	64.6	67.3	71.4	76.5	80.4	82.5	82.4	78.4	74.4	67.9	64.0
		MCDB	65.2	69.6	75.3	80.7	85.5	89.5	91.7	92.0	88.4	83.5	73.6	66.8
	2%	WB	59.6	60.4	64.4	68.9	74.5	78.7	81.0	80.5	76.7	71.5	64.2	59.7
		MCDB	63.1	64.6	71.7	77.1	83.2	88.8	90.2	89.7	86.3	79.4	68.5	63.0
	5%	WB	54.3	57.0	61.5	67.0	72.9	77.5	79.9	78.9	75.1	68.9	61.3	55.1
		MCDB	56.5	60.7	67.7	74.5	81.4	87.1	89.0	87.7	83.9	75.4	66.2	57.9
	10%	WB	47.9	51.4	58.2	64.9	71.2	76.1	78.6	77.5	73.4	66.2	57.6	50.7
		MCDB	52.1	56.6	63.5	71.6	79.3	85.1	87.2	85.9	81.0	72.9	62.7	54.0
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.8	19.4	20.8	19.6	19.5	18.0	19.3	22.2	23.2	20.2	15.9
		MCDBR	21.5	23.8	24.4	24.0	22.2	21.8	19.7	22.1	25.2	27.4	25.5	20.2
		MCWBR	17.5	17.7	15.6	12.6	9.8	8.4	8.0	9.5	11.3	13.8	16.9	16.6
	5% WB	MCDBR	18.4	21.0	21.3	19.8	19.0	19.5	18.4	19.4	21.5	22.5	21.1	17.5
		MCWBR	17.3	17.8	15.0	12.5	9.6	8.8	8.4	9.4	10.5	13.0	17.1	16.3
Clear-Sky Solar Irradiance	taub	0.312	0.322	0.354	0.396	0.433	0.452	0.485	0.477	0.412	0.352	0.321	0.307	
	taud	2.485	2.446	2.403	2.282	2.239	2.195	2.142	2.153	2.321	2.474	2.511	2.508	
	Ebn at Noon	279	290	289	281	271	265	255	254	266	275	274	274	
	Edh at Noon	26	30	35	42	44	46	48	47	37	29	25	24	
All-Sky Solar Radiation	RadAvg	728	916	1257	1616	1807	2060	1985	1808	1529	1154	798	614	
	RadStd	69	117	105	106	149	133	105	102	93	128	85	64	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page